

Polynomial identities



- 1 Follow the steps below to prove that $(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (bc + ad)^2$.
 - a Distribute the left-hand side of the equation.
 - b Distribute the right-hand side of the equation.
- 2 Prove $(m - n)^2 = m^2 - 2mn + n^2$.
- 3 Use the identity $(u + v)^2 = u^2 + 2uv + v^2$ with $v < 10$ to find the value of 1002^2 .
- 4 Prove $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$.
- 5 Use the identity $m^4 - n^4 = (m - n)(m + n)(m^2 + n^2)$ to find the value of $8^4 - 2^4$.
- 6 Use the identity $u^6 + v^6 = (u^3 - 2uv^2)^2 + (v^3 - 2u^2v)^2$ to find the value of $5^6 + 2^6$.
- 7 Consider the difference of two squares expression $u^2 - v^2$.
 - a Express the difference of two squares as a product.
 - b Use the identity $u^2 - v^2 = (u + v)(u - v)$ to find the value of $77^2 - 23^2$.
 - c Use the identity $u^2 - v^2 = (u + v)(u - v)$ to find the value of 24×16 .
- 8 Prove $x^5 + y^5 = (x + y)(x^4 - x^3y + x^2y^2 - xy^3 + y^4)$.
- 9 Prove $m^6 + n^6 = (m^3 - 2mn^2)^2 + (n^3 - 2m^2n)^2$.
- 10 Use the identity $(a^2 + b^2)(x^2 + y^2) = (ax - by)^2 + (bx + ay)^2$ to find the value of $(27^2 + 11^2)(42^2 + 94^2)$.



Applications

- 11** Kathleen noticed that if she multiplies the square number 289 by another square number 64, the result is a square number.
- a** What is the square root of the resulting product Kathleen found?
 - b** Let x^2 be a square number, and y^2 be different square number. Show that the product of two square numbers is always a square number.
- 12** Consider the sum of consecutive odd numbers.
- a** Evaluate the sum $1 + 3 + 5 + 7$.
 - b** Evaluate the sum of the first 7 odd numbers.
 - c** Form a simplified expression for the sum of the first t odd numbers:
 $1 + 3 + 5 + \dots + (2t - 1)$.
 - d** Using this result, find the sum of the first 70 odd numbers.
 - e** Can 3972 be the sum of consecutive odd numbers (starting from 1)?
- 13** Let mn be any two-digit number, where m is the tens digit and n is the unit digit.
- a** Express the value of the two-digit number in terms of m and n .
 - b** The digits are switched to form a new two-digit number. Express the value of the new two digit number in terms of m and n .
 - c** Show that the difference between the original two-digit number and the new two-digit number is always a multiple of 9.
- 14** Consider the product of three consecutive integers.
- a** 4080 is the product of three consecutive integers. Find the three integers using trial and error.
 - b** The product of any three consecutive integers in which x is the first integer is given by $x(x + 1)(x + 2)$. Which of the following numbers will the product always be divisible by?
- A 5 B 8 C 3 D 6**

15 Consider the following.



- a Expand and simplify $(m^2 + n^2)^2$.
 - b Prove that $(m^2 - n^2)^2 + (2mn)^2 = (m^2 + n^2)^2$
 - c What special relationship is this related to?
 - d Using the identity proven, find the other two integer length sides of a right-angled triangle in which one shorter side measures 30 units.
- 16 The identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ can be used to generate Pythagorean triples. Find the side lengths of a right-angled triangle using $x = 8$ and $y = 6$.
- 17 Consider the expression $(x - a)(x^2 + ax + a^2)$.
- a Distribute and simplify the expression.
 - b Show that 999 973 is not prime by expressing it as the product of two factors.
- 18 Consider the expression $(2x^2 + x + 2)(x^2 + 2x + 2)$.
- a Distribute and simplify the expression.
 - b Now, evaluate 212×122 .
 - c If the identity from part (a) is to be used to evaluate the products of three digit numbers, what value does x represent?

19 Consider the following pattern:

$$4^2 = 3^2 + 3 + 4$$

$$5^2 = 4^2 + 4 + 5$$

$$6^2 = 5^2 + 5 + 6$$

We wish to show that this pattern will continue for any square number. Let x^2 be the square number.

- a According to the pattern, how can x^2 be rewritten?
- b Prove that $x^2 = (x - 1)^2 + (x - 1) + x$.
- c Using the identity proven in the previous part, evaluate 101^2 .

20 Consider the expression $2 \times 3 \times 4 \times 5 + 1$.



- a Evaluate $2 \times 3 \times 4 \times 5 + 1$.
- b Find the missing value:

$$2 \times 3 \times 4 \times 5 + 1 = (\text{ })^2$$

- c We want to show that 1 more than the product of four consecutive integers is always a perfect square. Let x be the first of any four consecutive integers. Complete the following work to show that $x(x+1)(x+2)(x+3) + 1$ can be expressed as the square of a value:

$$\begin{aligned} x(x+1)(x+2)(x+3) + 1 &= (x^2 + x)(x^2 + \text{ }x + \text{ }) + 1 \\ &= x^4 + \text{ }x^3 + \text{ }x^2 + \text{ }x + \text{ } \\ &= (x^2 + \text{ }x + 1)(x^2 + \text{ }x + 1) \\ &= (x^2 + \text{ }x + 1)^{\text{ }} \end{aligned}$$